

Normal Form for Context free language Greibach Normal Form [GNF]

In GNF there is no restriction on the number of symbols on the right hand side but there is restriction on the terminals and variables appear on the right hand side of the production.

Definition: Let $G = (V, T, P, S)$ be a CFG, the CFG G is said to be in GNF if all the productions are of the form

$$A \rightarrow a\alpha$$

where $a \in T$ and $\alpha \in V^*$ i.e. the first symbol on the right hand side of the production must be a terminal & it can be followed by zero or more variables.

Simplification of Grammar

Simplification of context free grammar takes the following major steps are.

Step 1: Removal of useless symbol (non terminal)

Step 2: Removal of unit production

Step 3: Removal of ϵ production.

Step 1: Removal of useless symbol:

A symbol is termed useless if it is not useful i.e. it not generate any string.

Step 2: Removal of unit production

* A production of the form $A \rightarrow B$ is called unit production.

* To remove unit production, for each non terminal A , we compute A^+ , the set of non-terminal derived from A via unit production.

Removal of ϵ production

A symbol A is said to be nullable if $A \rightarrow \epsilon$. To remove ϵ production, we compute nullable set.

Two algorithms used to simplify CFG.

- a) TO find and remove unproductive
- b) TO find and remove unreachable variable

① Grammar $S \rightarrow AB|AC$

$A \rightarrow aAb|E$

$B \rightarrow aA$

$C \rightarrow bca$

$D \rightarrow AB$

→ remove unproductive
→ remove unreachable

Initial $UP = \{S, A, B, C, D\}$

Initial $P = \{a, b\}$

→ $P = \{a, b, A\}$

$A \rightarrow aAb|E$

→ $P = \{a, b, A, B\}$

$B \rightarrow aA$

$P = \{a, b, S, A, B\}$

$S \rightarrow AB$

$P = \{a, b, S, A, B, D\}$

$D \rightarrow AB$

C is unproductive

∴ drop RHS or LHS with unproductive

$S \rightarrow AB$

$A \rightarrow aAb|E$

$B \rightarrow aA$

$D \rightarrow AB$

$X \rightarrow \infty$
↑

all symbol
in RHS is
productive
LHS also
as productive

Remove unreachable

Initial Reachable = $\{S\}$

UR = $\{A, B, D\}$

[mark every nonterminal symbol as unreachable]

Reachable = $\{S, A, B\}$ $S \rightarrow AB$

$X \rightarrow \alpha \underline{A} \beta$
 $\downarrow$ reachable $\uparrow$ is yet not reachable
 Mark 'A' as reachable

D is unreachable, so

Remove D from both from

LHS, RHS

After removing useless grammar symbols

$\begin{cases} S \rightarrow AB \\ A \rightarrow aAB \mid \epsilon \\ B \rightarrow bA \end{cases}$

$S \rightarrow abs \mid abA \mid abB$

$A \rightarrow cd$

$B \rightarrow aB$

$C \rightarrow dc$

→ Remove unproductive
 → Remove unreachable

Initial UP = $\{S, A, B, C\}$

initial P = $\{a, b, c, d\}$

Loop — what are gets added as productive

$P = \{a, b, c, d, A\}$ $A \rightarrow cd$
 $P = \{a, b, c, d, A, C\}$ $C \rightarrow dc$
 $P = \{a, b, c, d, A, C, S\}$ $S \rightarrow abs$
 $S \rightarrow abA$

B is unproductive we remove from RHS,

$$S \rightarrow abs \mid abA$$

$$A \rightarrow cd$$

$$B \rightarrow dc$$

Remove unreachable symbols.

$$R = \{S\} \quad UR = \{A, C\}$$

loop: $R = \{S, A\}$

C is unreachable

$$\therefore S \rightarrow abs \mid abA$$

$$A \rightarrow \underline{cd}$$

left hand reaches
right hand

ex:

3) $S \rightarrow aab \mid abA \mid aAT$ Remove useless symbols
 $A \rightarrow aA$
 $B \rightarrow ab \mid b$
 $C \rightarrow ad$

1. Remove unproductive
2. Remove unreachable

1. Remove unproductive

$$\text{Initial } P = \{a, b, d\}$$

$$\text{initial } UP = \{S, A, B, C\}$$

loop:

$$P = \{a, b, d, c\} \quad C \rightarrow ad$$

$$P = \{a, b, d, c, B\} \quad B \rightarrow ab$$

$$P = \{a, b, d, c, B, S\}$$

2nd loop

3rd loop

unproductive are A & T

$$\therefore S \rightarrow aab$$

$$B \rightarrow ab|b$$

$$C \rightarrow ad$$

Remove unreachable

$$\text{Initial } R = \{S\}$$

$$R = \{S, B\}$$

$$\therefore UR = \{C\} \quad C \text{ is unreachable}$$

$$\therefore S \rightarrow aab$$

$$B \rightarrow ab|b$$

$$\underline{\underline{\quad\quad\quad}}$$

(4) Remove useless symbol / production

$$S \rightarrow AB|a$$

$$A \rightarrow Bclb$$

$$B \rightarrow aB|c$$

$$C \rightarrow ac|B$$

Remove unproductive

$$P = \{a, b\}$$

$$UP = \{S, A, B, C\}$$

$$P = \{a, b, S\} \quad S \rightarrow a$$

$$P = \{a, b, S, A\} \quad A \rightarrow B$$

$$\therefore \begin{array}{l} S \rightarrow a \\ A \rightarrow B \end{array} \quad \begin{array}{l} S \rightarrow a \\ A \rightarrow b \end{array}$$

2. Remove unreachable

$$R = \{S\}$$

UR =

$\therefore S \rightarrow a$ [A is unreachable]

5 $S \rightarrow AB|AC$

$$A \rightarrow aAb|bAa|a$$

$$B \rightarrow bBa|aAb|AB$$

$$C \rightarrow aBCA|aDb$$

$$D \rightarrow bD|ac.$$

Remove unproductive

$$P = \{a, b\}$$

$$UP = \{S, A, B, C, D\}$$

$$P = \{a, b, A\} \quad A \rightarrow a$$

$$P = \{a, b, A, B\} \quad B \rightarrow bBa$$

$$P = \{a, b, A, B, S\} \quad S \rightarrow AB$$

$$\therefore S \rightarrow AB$$

$$A \rightarrow aAb|bAa|a$$

$$B \rightarrow bBa|aAb|AB$$

$\therefore$ Remove C & D unproductive remove from grammar

$\Rightarrow$ Remove unreachable

$$R = \{S\}$$

$$UR = \{A, B\}$$

$$R = \{S, A, B\}$$

There are no unreachable symbols
 $\therefore$ there is no change

$$S \rightarrow AB$$

$$A \rightarrow aAb \mid bAa \mid a$$

$$B \rightarrow bBa \mid aAb \mid AB$$

Remove ϵ production

Eliminating ϵ rules

$\Rightarrow$ Nullable variable:

A variable x is nullable iff either

(i) There is a rule $x \rightarrow \epsilon$ or

(ii) There is a rule $x \rightarrow pqr$ and p, q, r are all nullable.

So compute N , the set of nullable variable as follows.

1. Set N to the set of variable that satisfy (i).

2. Until an entire pass is made without adding anything to N do evaluate all other variables with respect to (i).

If any variable satisfies (2) & is not in N . insert it.

modifiable variable: A rule is modifiable iff it is of the form

$P \rightarrow \alpha Q \beta$ for some nullable Q .

$$\begin{aligned} \textcircled{2} \quad & S \rightarrow aTa \\ & T \rightarrow ABC \\ & A \rightarrow aA \mid C \\ & B \rightarrow Bb \mid C \\ & C \rightarrow c \mid \epsilon \end{aligned}$$

Initial nullable $N = \{C\}$ $C \rightarrow \epsilon$

$$N = \{C, B, A\} \quad \begin{aligned} A &\rightarrow C \\ B &\rightarrow C \end{aligned}$$

$$N = \{C, B, A, T\} \quad T \rightarrow ABC$$

All nullable NT

$$S \rightarrow aTa \mid aa$$

$$T \rightarrow ABC \mid BC \mid AC \mid AB \mid A \mid C \mid B \quad \begin{cases} \text{one at a time} \\ \& 2 \text{ at a time} \end{cases}$$

$$A \rightarrow aA \mid a \mid C$$

$$B \rightarrow Bb \mid b \mid C$$

$$\underline{\underline{C \rightarrow c}}$$

{ remove ϵ production }

$$\begin{aligned} \textcircled{2} \quad & S \rightarrow aAga \\ & A \rightarrow B/a \\ & B \rightarrow d/e \\ & d \rightarrow eC/e \end{aligned}$$

Remove ϵ production

First find nullable

$$N = \{C\} \quad C \rightarrow \epsilon$$

$$N = \{B, C\} \quad B \rightarrow C$$

$$N = \{A, B, C\} \quad A \rightarrow B, B \rightarrow C, C \rightarrow \epsilon$$

$$\therefore S \rightarrow aAGa/aGa/aAa/aa$$

$$A \rightarrow B/a$$

$$B \rightarrow C/e$$

$$d \rightarrow eC/e$$

Removal of unit production

The unit productions are the productions in which one non-terminal gives another non-terminal.

$$\text{ex: } X \rightarrow Y \quad \& \quad S \rightarrow Y$$

Step 1: To remove $X \rightarrow Y$ add production $X \rightarrow a$ to grammar rule, $Y \rightarrow a$ occurs in the grammar whenever

Step 2: now delete $X \rightarrow X$ from grammar

Step 3: repeat step 1 & 2 until all production ~~reached~~ removed from grammar.

ex: $S \rightarrow 0A | 1B | C$

$A \rightarrow 0S | 00$

$B \rightarrow 1 | A$

$C \rightarrow 01$

$S \rightarrow 0A$ $A \rightarrow 0S$ $B \rightarrow 1$ $C \rightarrow 01$

$S \rightarrow 1B$ $A \rightarrow 00$ $B \rightarrow A$ — unit production

$S \rightarrow C$

$S \rightarrow$

$S \rightarrow 0A | 1B | 01$

$A \rightarrow 0S | 00$

$B \rightarrow 1 | 0S | 00$

$C \rightarrow 01$ //

Chomsky Normal Form [CNF]

CNF stands for Chomsky normal form.
A CFG is in CNF if all production rules satisfy one of the following conditions.

$\Rightarrow$ Start symbol generating ϵ
eg: $A \rightarrow \epsilon$

$\Rightarrow$ A non terminal generating two non terminal

ex: $S \rightarrow AB$

$\Rightarrow$ A non terminal generating terminal
eg: $S \rightarrow a$

Removal of unit production

A unit production is a production of the form $A \rightarrow B$ where both A & B are variable

1) Eliminate all unit production from the grammar,

$$S \rightarrow AB$$

$$A \rightarrow a$$

$$B \rightarrow C|b$$

$$C \rightarrow D$$

$$D \rightarrow E|bc$$

$$E \rightarrow d|Ab$$

Non unit production

$$S \rightarrow AB$$

$$A \rightarrow a$$

$$B \rightarrow b$$

$$D \rightarrow bc$$

Unit production

$$B \rightarrow C$$

$$C \rightarrow D$$

$$D \rightarrow E$$

It is clear from the depending graph that $D \rightarrow E$, so the non unit productions generated from 'E' can also be generated from D.

$$E \rightarrow d|Ab$$

Since

$$D \rightarrow E$$

$$D \rightarrow d|Ab|bc$$

$$C \rightarrow D \rightarrow E$$

$$C \rightarrow d|Ab|bc$$

Since $B \rightarrow C$, $C \rightarrow D$, $D \rightarrow E$

$$B \rightarrow b|d|Ab|bc$$

The final grammar obtained after eliminating

$$V = \{S, A, B, C, D, E\}$$

$$T = \{a, b, d\}$$

$$P = \{S \rightarrow AB$$

$$A \rightarrow a$$

$$B \rightarrow b|d|Ab|bc$$

$$C \rightarrow bc|d|Ab$$

$$D \rightarrow bc|d|Ab$$

$$E \rightarrow d|Ab$$

$\}$
S is the start symbol,,

② Eliminate unit productions from the grammar.

$$S \rightarrow AO|B$$

$$B \rightarrow A|11$$

$$A \rightarrow O|12|B$$

non unit production

$$S \rightarrow AO$$

$$B \rightarrow 11$$

$$A \rightarrow O|12$$

unit production

$$S \rightarrow B$$

$$B \rightarrow A$$

$$A \rightarrow B$$

It is clear from the dependence graph that
 $S \rightarrow B$, $B \rightarrow A$, & $A \rightarrow B$ so the new production
for S, A & B are

$$S \rightarrow AO|11|O|12$$

$$B \rightarrow O|12|11$$

$$A \rightarrow O|12|11$$

The resulting grammar without unit production.

$$V = \{S, A, B\}$$

$$T = \{O, 1, 2\}$$

$$P = \{ S \rightarrow A0111/0112$$

$$B \rightarrow 011211$$

$$A \rightarrow 011211 \}$$

S is start symbol

③ Eliminate unit production

$$S \rightarrow Aa/B/Ca$$

$$B \rightarrow aB/b$$

$$C \rightarrow Db/D$$

$$D \rightarrow E/d$$

$$E \rightarrow ab$$

unit production

$$\begin{cases} S \rightarrow B \\ C \rightarrow D \\ D \rightarrow E \end{cases}$$

non unit production

$$S \rightarrow Aa/Ca$$

$$B \rightarrow aB/b$$

$$C \rightarrow Db$$

$$D \rightarrow d$$

$$E \rightarrow ab$$

$$S \rightarrow B$$

$$S \rightarrow Aa/aB/b/Ca \quad \}$$

$$D \rightarrow E$$

$$D \rightarrow ab/d$$

$$C \rightarrow D \quad D \rightarrow E$$

$$C \rightarrow Db/abd$$

The resulting grammar is

$$V = \{ S, A, B, C, D, E \}$$

$$T = \{ a, b, d \}$$

$$P = \{ S \rightarrow Aa | aB | b | Ca$$

$$B \rightarrow aB | b$$

$$C \rightarrow Db | ab | d$$

$$D \rightarrow d | ab$$

$$E \rightarrow ab \}$$

S is start symbol.

Normal Form For Context Free Language Greibach Normal Form [GNF]

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Converting CFG to CNF

A context free grammar $G = (V, \Sigma, R, S)$ is said to be in Chomsky Normal Form (CNF), iff every rule is of one of the following forms.

$X \rightarrow a$ where $a \in \Sigma$ or

$X \rightarrow BC$ where $B, C \in V - \Sigma$

ex: $S \rightarrow AB, A \rightarrow a, B \rightarrow b$

CNF [Greibach Normal Form]

CNF is a context free grammar $G = (V, \Sigma, R, S)$ where all rules have one of the following forms:

$X \rightarrow a\beta$ where $a \in \Sigma$ & $\beta \in (V - \Sigma)^*$

ex: $S \rightarrow aA \mid aAB, A \rightarrow, B \rightarrow b$

conversion takes 4 steps [CFG to CNF]

There exist 4-steps algorithm to convert a CFG G into a new grammar G_c such that: $L(G) = L(G_c) - \{\epsilon\}$

1. $G^I =$ remove EPS $S \rightarrow \epsilon$

2. $G^{II} =$ remove unit $A \rightarrow B$

3. $G^{III} =$ remove mixed $A \rightarrow AB$

4. $G^{IV} =$ remove long $S \rightarrow ABCD$

return G_c

① convert the given grammar into CNF

$$G = S \rightarrow aACa / aCa / aAa / aa$$

$$A \rightarrow eC / e / a$$

$$B \rightarrow eC / e$$

$$C \rightarrow e / eC$$

$$S \rightarrow aACa$$

$$A \rightarrow B / a$$

$$B \rightarrow C / e$$

$$C \rightarrow eC / e$$

Step 1: Remove ϵ production

$$N = \{C\} \quad C \rightarrow \epsilon$$

$$N = \{B, C\} \quad B \rightarrow C \quad C \rightarrow \epsilon$$

$$N = \{A, B, C\} \quad A \rightarrow B \rightarrow C \quad C \rightarrow \epsilon$$

$$G' = \left\{ \begin{array}{l} S \rightarrow aACa / aCa / aAa / aa \\ A \rightarrow B / a \\ B \rightarrow C / e \\ C \rightarrow eC / e \end{array} \right\}$$

Step 2: Remove unit productions

$$G'' = \left\{ \begin{array}{l} S \rightarrow aACa / aCa / aAa / aa \\ A \rightarrow eC / e / a \\ B \rightarrow eC / e \\ C \rightarrow eC / e \end{array} \right\}$$

$$\begin{array}{l} B \rightarrow C \\ C \rightarrow eC / e \end{array}$$

Step 3: Remove mixed symbol

$$G^{III} = \left\{ \begin{array}{l} S \rightarrow TaAcTa \mid TaCTa \mid TaATa \mid TaTa \\ Ta \rightarrow a \\ A \rightarrow TeC \mid e \mid a \\ Te \rightarrow e \\ B \rightarrow TeC \mid e \\ C \rightarrow TeC \mid e \end{array} \right.$$

Step 4: Remove Long production

$$G^{IV} = \left\{ \begin{array}{l} S \rightarrow TaS_1 \mid TaS_2 \mid TaS_3 \mid TaTa \\ S_1 \rightarrow AS_2 \\ S_2 \rightarrow cTa \\ S_3 \rightarrow ATa \\ Ta \rightarrow a \\ A \rightarrow TeC \mid e \mid a \\ Te \rightarrow e \\ B \rightarrow TeC \mid e \\ C \rightarrow TeC \mid e \end{array} \right. \left\{ \begin{array}{l} S \rightarrow Ta \underline{AcTa} \\ S \rightarrow Ta S_1 \\ S_1 \rightarrow A \underline{CTa} \\ \quad \quad \quad S_2 \end{array} \right.$$

Q) convert the following grammar to CNF

$$G = \{ S \rightarrow aSa | B \\ B \rightarrow bbC | bb \\ C \rightarrow cC | \epsilon \}$$

CNF form [Chomsky Normal Form]

$$A \rightarrow a$$

$$A \rightarrow BC$$

NT - terminal

$$NT \rightarrow NTNT \quad \{ NT : \text{nonterminal} \}$$

Step 1: Remove ϵ production.

$$\text{Nullable} = \{ C \} \quad C \rightarrow \epsilon$$

$$S \rightarrow aSa | B$$

$$B \rightarrow bbC | bb$$

$$C \rightarrow cC | c$$

Step 2: Remove unit production.

$$\{ \begin{array}{l} NT \rightarrow NT \\ A \rightarrow B \end{array} \}$$

$$S \rightarrow aSa | bbC | bb$$

$$B \rightarrow bbC | bb$$

$$C \rightarrow cC | c$$

Step 3: Remove mixed production

$$\{ NT \rightarrow \gamma NT \}$$

$$S \rightarrow T_a S T_a | T_b T_b C | T_b T_b$$

$$B \rightarrow T_b T_b C | T_b T_b$$

$$C \rightarrow T_c C | c$$

⑤

Step 4: Remove long production

$$\left\{ \begin{array}{l} NF \rightarrow T \\ NT \rightarrow NTNT \end{array} \right\}$$

$$S \rightarrow T_a S_1 \mid T_b S_2 \mid T_b T_b$$

$$S_1 \rightarrow S T_a$$

$$S_2 \rightarrow T_b C$$

$$B \rightarrow T_b S_2 \mid T_b T_b$$

$$C \rightarrow T_c C \mid c$$

$$T_a \rightarrow a$$

$$T_b \rightarrow b$$

$$T_c \rightarrow c$$

③ convert given CFG to CNF

$$S \rightarrow ABC$$

$$A \rightarrow AC \mid D$$

$$B \rightarrow bB \mid A \mid \epsilon$$

$$C \rightarrow Ac \mid Cc \mid \epsilon$$

$$D \rightarrow aa$$

Step 1: Remove ϵ production

$$\text{Nullable } \{C\} \quad C \rightarrow \epsilon$$

$$\text{Nullable } \{C, B\} \quad B \rightarrow \epsilon$$

$$S \rightarrow ABC \mid AC \mid AB \mid A$$

$$A \rightarrow AC \mid a \mid D$$

$$B \rightarrow bB \mid b \mid A$$

$$C \rightarrow Ac \mid Cc \mid c$$

$$D \rightarrow aa$$

Step 2 Remove unit production

$$\begin{cases} S \rightarrow A \\ B \rightarrow A \\ A \rightarrow D \end{cases}$$

$$S \rightarrow ABC | AC | AB | aG | a | aa$$

$$A \rightarrow aG | a | aa$$

$$B \rightarrow bB | b | aG | a | aa$$

$$G \rightarrow Ac | Cc | c$$

$$D \rightarrow aa$$

Step 3: Remove mixed production

$$S \rightarrow ABC | AC | AB | TaG | \cancel{a} | TaTa$$

$$Ta \rightarrow a$$

$$A \rightarrow TaG | a | TaTa$$

$$B \rightarrow TbB | b | TaG | a | TaTa$$

$$Tb \rightarrow b$$

$$G \rightarrow ATc | cTc | c$$

$$Tc \rightarrow c$$

$$D \rightarrow aa$$

Step 4: Remove long production

$$S \rightarrow AS_1 | AC | AB | TaG | a | TaTa$$

$$S_1 \rightarrow BC$$

$$Ta \rightarrow a$$

$$A \rightarrow TaG | a | TaTa$$

$$B \rightarrow TbB | b | TaG | a | TaTa$$

$$Tb \rightarrow b$$

$$G \rightarrow ATc | cTc | c$$

$$Tc \rightarrow c$$

$$D \rightarrow aa$$

problems on GNF

- convert the given grammar CNF
- Eliminate left recursion
- convert into GNF.

Conversion Rules

1. obtain grammar in CNF
2. Rename variables
3. Use substitution method to obtain production to the form

$$A_i \rightarrow A_j \text{ where } i < j$$

4. Remove Left Recursion

Apply 3 and 4 until grammar is in GNF.

- ① convert the following grammar to GNF form.

~~$$S \rightarrow aBc \quad B \rightarrow b$$~~

1. Introduce new symbol nonterminal C

$$S \rightarrow aBC$$

$$B \rightarrow b$$

$$C \rightarrow c$$

- ② $S \rightarrow Bc$ — $\{NT, t\}$ it is not in GNF
 $B \rightarrow b$

$$S \rightarrow bc \quad \rightarrow S \rightarrow bC \quad \left. \begin{array}{l} c \rightarrow c \end{array} \right\} \text{ removed } B \rightarrow b$$

5) convert the following grammar in CNF form.

$$S \rightarrow AA \mid 0$$

$$A \rightarrow SS \mid 1$$

$$\begin{cases} A \rightarrow BC \\ A \rightarrow a \\ \text{CNF form} \end{cases}$$

1) This grammar is in CNF

$$A = A_2$$

2) Renaming $S = A_1$

$$A_1 \rightarrow A_2 A_2 \mid 0 \quad \text{--- (a)}$$

$$A_2 \rightarrow A_1 A_1 \mid 1$$

3) $A_i \rightarrow A_j \quad i < j$
so use substitution rule $A_2 \rightarrow A_2 A_2 A_1 \mid 0 A_1 \mid 1$

left recursion Remove left recursion

$$A \rightarrow A\alpha_1 \mid A\alpha_2 \dots \mid A\alpha_n \mid \beta_1 \mid \beta_2 \dots \mid \beta_m$$

Left recursion A's

$$\text{i.e. } A \rightarrow \beta_1 \cancel{A} \mid \beta_2 \cancel{A} \dots \mid \beta_m \cancel{A}$$

{ $\cancel{A}$ new symbol?}

$$\cancel{A} \rightarrow \alpha_1 \cancel{A} \mid \alpha_2 \cancel{A} \dots \mid \alpha_n \cancel{A} \mid \epsilon$$

$$A_2 \rightarrow 0 A_1 \cancel{X} \mid 1 \cancel{X}$$

$$\cancel{X} \rightarrow A_2 A_1 \cancel{X} \mid \epsilon$$

{instead of A' replace with X}

Rewrite $\Rightarrow A_2 \rightarrow 0 A_1 \mid 1 \mid 0 A_1 \cancel{X} \mid 1 \cancel{X}$ $\rightarrow$ CNF $\checkmark$ --- (b)
remove ϵ $\cancel{X} \rightarrow A_2 A_1 \cancel{X} \mid A_2 A_1$ --- (c)

production (a) $A_1 \rightarrow A_2 A_2 \mid 0$

Not in CNF

(b) $A_2 \rightarrow 0 A_1 \mid 1 \mid 0 A_1 \cancel{X} \mid 1 \cancel{X}$ --- in CNF

(c) $\cancel{X} \rightarrow A_2 A_1 \cancel{X} \mid A_2 A_1$ --- Not in CNF

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(a) substitute A_2 in RHS

$$A_1 = 0A_1A_2 \mid 1A_2 \mid 0A_1 \bar{A}_2 \mid 1\bar{A}_2 \mid 0 \quad - \text{in QNF}$$

③ substitute A2

$$x \rightarrow 0A_1A_1x \mid 1A_1x \mid 0A_1xA_1x \mid 1xA_1x$$

$$|0A, A\rangle, |1A\rangle, |0A, XA\rangle, |1XA\rangle \quad - \text{in } \underline{\underline{QNF}}$$

② convert the following grammar to CNF

$$A \rightarrow BC$$

$$B \rightarrow CA \mid b$$

$$C \rightarrow AB|a$$

① All production are in CNF

② $A = A_1, B = A_2, C = A_3$

$$A_1 \rightarrow A_2 A_3 \quad A_2 \rightarrow A_3 A_1 / b \quad A_3 \rightarrow A_1 A_2 / a$$

(3) $A_i \rightarrow A_j, \quad i < j$

↑ not in formal
∴ 371

Substitution

$$A_3 \rightarrow A_1 A_2 / a \quad - \text{substitute for } A_1$$

$$A_3 \rightarrow A_2 A_3 A_2 / a \quad \text{is still } \exists \gamma \geq 2$$

$L \rightarrow$ substitute for A_2

Rewrite

$A_3 \rightarrow A_3 A_1 \underbrace{A_3 A_2}_\alpha \mid \underbrace{b A_3 A_2}_\beta \mid \underbrace{a}_{\beta_2}$ it has left recursion

Formula for remove left recursion

$$\left\{ \begin{array}{l} A \rightarrow A\alpha_1 \mid \beta_1 \mid \beta_2 \\ A \rightarrow \beta_1 x \mid \beta_2 x \\ x \rightarrow \alpha_1 x \mid \epsilon \end{array} \right\}$$

$$\left\{ \begin{array}{l} A_3 \rightarrow bA_3A_2x \mid Ax \\ x \rightarrow A_1A_3A_2x \mid \epsilon \end{array} \right\} \text{ Remove } \epsilon$$

$$\textcircled{1} A_3 \rightarrow bA_3A_2x \mid Ax \mid bA_3A_2a$$

$$\textcircled{2} x \rightarrow A_1A_3A_2x \mid A_1A_3A_2$$

$$\textcircled{3} A_1 \rightarrow A_2A_3 \quad * \quad \text{not in GNF}$$

$$\textcircled{4} A_2 \rightarrow A_3A_1 \mid b \quad * \quad \text{not in GNF}$$

$$\textcircled{5} A_3 \rightarrow bA_3A_2x \mid Ax \mid bA_3A_2a \quad - \text{GNF}$$

$$\textcircled{6} x \rightarrow A_1A_3A_2x \mid A_1A_3A_2 \quad \text{not in GNF}$$

Rewrite 2 by substituting for A_3

$$A_2 \rightarrow bA_3A_2xA_1 \mid Ax A_1 \mid bA_3A_2A_1 \mid aA_1 \mid b$$

↳ in GNF

Rewrite 1 by substituting for A_2

$$A_1 \rightarrow bA_3A_2xA_1A_3 \mid Ax A_1A_3 \mid bA_3A_2A_1A_3 \mid aA_1A_3 \mid bA_3$$

↳ in GNF

④ $X \rightarrow bA_3A_2XA_1A_3 - A_3A_2X \mid$
 $aXA_1A_3 - A_3A_2X \mid$
 $bA_3A_2A_1A_3 - A_3A_2X \mid$
 $aA_1A_3 - A_3A_2X \mid$
 $bA_3 - A_3A_2X \mid$

$bA_3A_2XA_1A_3 \mid A_3A_2 \mid$
 $aXA_1A_3 \mid A_3A_2 \mid$
 $bA_3A_2A_1A_3 \mid A_3A_2 \mid$
 $aA_1A_3 \mid A_3A_2 \mid$
 $bA_3 \mid A_3A_2 \mid$

$\rightarrow$ in GNF

} substitute A_1

pumping lemma for context free language

③ convert into GNF

$$S \rightarrow CA \mid BB$$

$$B \rightarrow b \mid SB$$

$$C \rightarrow b$$

$$A \rightarrow a$$

① This grammar is in CNF

② Renaming $S = A_1, C = A_2, A = A_3, B = A_4$

$$A_1 \rightarrow A_2 A_3 \mid A_4 A_4$$

$$A_4 \rightarrow b \mid A_1 A_4$$

$$A_2 \rightarrow b$$

$$A_3 \rightarrow a$$

③ $A_i \rightarrow A_j k$ $\underline{i < j}$ & should never be $i > j$

$$A_1 \rightarrow A_2 A_3 \mid A_4 A_4$$

$$A_4 \rightarrow b \mid A_1 A_4$$

$$A_2 \rightarrow b$$

$$A_3 \rightarrow a$$

Resolve $A_4 \rightarrow \underline{A_1} A_4$, replace A_1

$$A_4 \rightarrow b \mid \underline{A_2} A_3 A_4 \mid A_4 A_4 A_4 \quad \text{replace } A_2$$

$$A_4 \rightarrow b \mid b A_3 A_4 \mid \underline{A_4 A_4 A_4}$$

↓
left recursion

Remove left recursion Introduce new variable.

$$Z \rightarrow A_4 A_4 Z \mid A_4 A_4$$

$$A_4 \rightarrow b \mid b A_3 A_4 \mid b Z \mid b A_3 A_4 Z$$

$$Z \rightarrow A_4 A_4 Z \mid A_4 A_4$$

$$\left\{ \begin{array}{l} A A A \\ A A C \mid A A \\ A_4 = b \mid b A_3 A_4 \mid b Z \\ A_4 \rightarrow b \mid A_3 A_4 \\ A_4 \rightarrow A_4 A_4 A_4 \end{array} \right.$$

new grammar

$$A_1 \rightarrow A_2 A_3 \mid A_4 A_4$$

$$A_4 \rightarrow b \mid b A_3 A_4 \mid b Z \mid b A_3 A_4 Z$$

$$Z \rightarrow A_4 A_4 \mid A_4 A_4 Z$$

$$A_2 \rightarrow b$$

$$A_3 \rightarrow a$$

Replace A_2 & A_4

$$A_1 \rightarrow b A_3 \mid b A_4 \mid b A_3 A_4 A_4 \mid b Z A_4 \mid b A_3 A_4 Z A_4$$

$$A_4 \rightarrow b \mid b A_3 A_4 \mid b Z \mid b A_3 A_4 Z$$

$$Z \rightarrow b A_4 \mid b A_3 A_4 \mid b Z A_4 \mid$$

$$b A_3 A_4 Z A_4 \mid$$

$$b A_4 Z \mid b A_3 A_4 A_4 Z \mid b Z A_4 Z \mid b A_3 A_4 Z A_4 Z$$

$$A_2 \rightarrow b$$

$$A_3 \rightarrow a$$

$\therefore$ It is in CNF //

pumping Lemma for context free language

pumping Lemma for CFL is used to prove that a language is not a context free

Assume L is a context free language, there is a pumping length n such that any string $w \in L$ of length $\geq n$ can be written as

$$|w| \geq n$$

we can break w into 5 string
 $w = uvxyz$ such that

$$|vxy| \leq n$$

$$|vy| \neq \epsilon$$

$\forall k \geq 0$ the string $uv^kxy^kz \in L$

Note: select k such that the resulting string is not in L .

prove that a language is not CF using pumping lemma CFL follows the steps (we prove using contradiction)

- ① Assume that L is context free
- ② the pumping length say n
- ③ All strings longer than n can be pumped
 $|w| \geq n$
- ④ Now find a string ' w ' in L such that
 $|w| \geq n$
- ⑤ Divide w into ~~xyz~~ $uvxyz$
- ⑥ show that $uv^kxy^kz \notin L$ for some k
- ⑦ then consider the ways that w can be divided into $uvxyz$
- ⑧ show that none of these can satisfy all the 3 pumping conditions at same time.
- ⑨ w cannot be pumped (contradiction)

① find out whether $L = \{x^n y^n z^n \mid n \geq 1\}$ is context free or not

Let L is context free (contradiction)

$\Rightarrow$ Now we can take a string such that
 $S = x^n y^n z^n$

$\Rightarrow$ we divide S into 5 parts $u v x y z$

case (i): $n=4$ so, $S \rightarrow x^4 y^4 z^4$

v & y each contain only one type of symbols.

$$\begin{array}{ccccccc} x & x & x & x & y & y & y & y & z & z & z & z \\ | & | & | & | & & & & & | & | & | & | \\ u & v & & x & & & & & y & z & & \end{array}$$
 $\Rightarrow$ divide into 5 parts
 $u v x y z$

$u v^k x y^k z$ (Let $k=2$)

$\Rightarrow u v^2 x y^2 z$

$\Rightarrow x x x x x x y y y y z z z z z$

$\Rightarrow x^6 y^4 z^5$

$x^6 y^4 z^5 \notin L$

$u = z$
 $v = xx$
 $x = xy y y y z$
 $y = z$
 $z = zz$

$\therefore$ the given language is not context free.

case (ii): either x or y has more than one kind of symbol

$$S = x^n y^n z^n$$

$$n=4 \text{ so } x^4 y^4 z^4$$

$$\Rightarrow \underbrace{xxxx}_{u} \underbrace{yyyy}_{v} \underbrace{zzzz}_{z}$$

$\Rightarrow$ let assume $k=2$

$$\Rightarrow uv^k xy^k z$$

$$\Rightarrow uv^2 xy^2 z$$

$$\Rightarrow xxxxyyxxxyyyzzzz$$

$$\Rightarrow x^4 y^2 x^2 y^5 z^4 \notin L$$

$\Rightarrow$ 5 parts
 $\downarrow$
 $u \cancel{xy} z$

$$u = xx$$

$$v = xxxy$$

$$x = y$$

$$y = y$$

$$z = zzzz$$

② $L = \{a^p \mid p \text{ is a prime}\}$ is not context free

If L is context free, we can apply pumping lemma

Let n be the constant of the lemma

consider sentence a^p , where $p > n$.

$$\text{let } |vz| = k \leq n$$

$$\text{then } v^k = vz = a^k$$

Then a^p can be written as
 $a^{p-k} a^k$

If vy is pumped $(p-k)$ times, we get a sentence

$$a^{p-k} a^{k(p-k)} = a^{(p-k)(k+1)}$$

→ see solution in next page

It is not a prime,

∴ $\{a^p \mid p \text{ is a prime}\}$ is not context free.

③ show that $L = \{ww \mid w \in \{0,1\}^*\}$ is not context free.

Assume L is context free
 L must have a pumping length (say p)
 Now take string S such that

$$S = 0^p 1^p 0^p 1^p$$

divide string into 5 parts $uvxyz$
 such that $|vxy| \leq p$, $|vy| \neq \epsilon$

eg take $p = 5$

$$\therefore 0^5 1^5 0^5 1^5$$

$$\underbrace{0000001}_{u} \underbrace{1111}_{vxy} \underbrace{000001111}_{z}$$

$$uv^2xy^2z \quad \text{Let } k=2$$

$$000002111111000001111$$

$$\therefore 0^5 1^7 0^5 1^5 \in L$$

The given language is not context free.

$$a^{p-k} a^{k(p-k)}$$

$$a^{p-k} a^{kp-k^2}$$

$$a^{p-k+kp-k^2}$$

$$a^{p-k+kp-k^2}$$

$$a^{(p+kp)-(k+k^2)}$$

$$a^{(p+kp)-(k+k^2)}$$

$$a^{p(1+k)-k(1+k)}$$

$$a^{(p+k)(p-k)}$$

$$\underline{\underline{a^{(p+k)(p-k)}}}$$

4) $L = \{a^n b^n c^n : n \geq 0\}$ is not CFL

Assume L is context free

Let $w = a^k b^k c^k$

* $|w| \geq k$ we divide

String $uvxy$

$$|vxy| \leq k$$

$$|vy| \neq 0 \quad \text{At } q \geq p, uv^q xy^q z$$

Let $k=4$

$$w = a^4 b^4 c^4$$

$$\begin{array}{cccccccc} a & a & a & a & b & b & b & b & c & c & c & c \\ \hline u & v & x & y & z & & & & & & & \end{array}$$

$$|vxy| \leq 4$$

$$aaa \leq 4$$

$$3 \leq 4$$

$$|vy| \neq 0$$

$$2 \neq 0$$

Let $q=2$

$$uv^2 xy^2 z$$

$$a(a)^2 a(a)^2 z$$

$$aaaaaaabbbbbbcccc$$

$$a^6 b^4 c^4 \notin L$$

$\therefore$ not CFL

⑤ $L = \{a^{n^2} : n \geq 0\}$ is not CFL

Assume L is context free

Let $w = a^k$

* $|w| \geq k$ we divide string into $uvxyz$

$$|vxy| \leq k$$

$$|vy| \neq 0 \quad \forall q \geq 0 \quad uv^qxy^qz$$

Let $k=2$

$$w = a^{2^2}$$

$$= a a a a$$

$$\begin{array}{cccc} \downarrow & \downarrow & \downarrow & \downarrow \\ u & v & x & y & z \end{array}$$

$$|vxy| \leq 2$$

$$a a \leq 2$$

$$2 \leq 4 \quad \checkmark$$

Let $q=2$

$$uv^2xy^2z$$

$$a a a a a a \notin L$$

$\therefore$ not CFL

⑥ $L = \{wcw : w \in \{a,b\}^*\}$ is not CFL

Assume L is context free

Let $w = a^r b^k c a^r b^k$

* $|w| \geq k$ we divide $w = uvxyz$

$$|vxy| \leq k$$

$$|vy| \neq 0 \quad \forall q \geq 0 \quad uv^qxy^qz$$

Let $k=4$

$$a^4 b^4 c a^4 b^4$$

$$w = a a a a b b b b c a a a a b b b b$$

$$\begin{array}{cccc} \downarrow & \downarrow & \downarrow & \downarrow \\ u & v & x & y & z \end{array}$$

$$|vxy| \leq 4$$

$$a a a \leq 4$$

$$3 \leq 4 \quad \checkmark$$

$$\rightarrow |vy| \neq 0$$

Let $q=2$

$$uv^2xy^2z$$

$$a(a)^2 a(a)^2 c$$

$$a^6 b^4 c a^4 b^4$$

$$\notin L$$

$\therefore$ not CFL

Closure properties of CFL.

1) union, concatenation, Kleen star

If L_1 & L_2 are context free languages. & then prove $L_1 \cup L_2$, $L_1 L_2$ and L_1^*

$$(i) G_1 = (V_1, T_1, P_1, S_1)$$

$$G_2 = (V_2, T_2, P_2, S_2)$$

$$G_3 = (V_1 \cup V_2 \cup S_3, T_1 \cup T_2, P_3, S_3)$$

S_3 is a start state G_3 and $S_3 \in (V_1 \cup V_2)$

$$P_3 = P_1 \cup P_2 \cup \{S_3 \rightarrow s_1 / s_2\}$$

$$L_3 = L_1 \cup L_2$$

$$(ii) G_4 = (V_1 \cup V_2 \cup S_4, T_1 \cup T_2, P_4, S_4)$$

S_4 is a start symbol for the grammar G_4 and $S_4 \in (V_1 \cup V_2)$

$$P_4 = P_1 \cup P_2 \cup \{S_4 \rightarrow s_1 s_2\}$$

$$\therefore L_3 = L_1 \cdot L_2$$

$$(iii) G_5 = (V, \cup S_5, T_1, P_5, S_5)$$

S_5 is the start symbol of grammar G_5

$$P_5 = P_1 \cup \{S_5 \rightarrow s_1 s_5 \mid \epsilon\}$$

$$TL_5 = L_s^*$$